

1. Basic Information

Title:

Authors:

Affiliation (University/Company):

Year:

Venue (Journal/Conference):

Paper Link (PDF / DOI / ArXiv):

2. Research Context

Domain:

- ☐ Machine Learning ☐ Deep Learning ☐ Computer Vision ☐ NLP ☐ IoT
☐ Cyber Security ☐ Data Mining ☐ Cloud/Edge AI ☐ Other: _____

Sub-domain (more specific):

(e.g., Object Detection, Sentiment Analysis, Intrusion Detection)

3. Problem Definition

Problem Statement:

- What exact problem are the authors trying to solve?
- Write it in **1–2 simple sentences** in your own words.

Why is this problem important?

- **Practical impact:** (real-world use, industry, society)
 - **Theoretical impact:** (new method, improved understanding, algorithmic novelty)
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4. Key Idea & Methodology

Core Idea / Contribution:

- What is the **main novelty** of the paper?
- What makes it different from previous work?

Model / Approach Used:

- Algorithms / Models (e.g., CNN, Transformer, LSTM, Random Forest)
- Architecture details (if any)
- Pipeline / workflow (input → processing → output)

Baseline Methods Compared Against:

- What existing methods are used for comparison?
-

5. Data & Experimental Setup

Datasets Used:

- Name
- Source
- Size
- Public or private?

Evaluation Metrics:

- Accuracy, F1, Precision, Recall, IoU, BLEU, mAP, RMSE, etc.

Experimental Settings:

- Train/validation/test split

- Hardware (GPU/CPU if mentioned)
 - Software/framework (PyTorch, TensorFlow, etc.)
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6. Results & Findings

Main Results:

- Best performance achieved
- Comparison with baselines (who wins and by how much)

Conclusion (Author's Claim):

- What do the authors finally conclude?
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7. Strengths

- Technical strengths
 - Clear novelty
 - Strong results
 - Good dataset choice
 - Real-world applicability
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8. Weaknesses & Limitations

- Methodological issues

- Dataset limitations
 - Overfitting / lack of generalization
 - High computational cost
 - Unrealistic assumptions
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9. Critical Analysis (Very Important for Thesis)

Unexplored Data:

- What type of data is missing?
- Different modality, language, environment, scale?

Unstudied Region / Scenario:

- Geographic, demographic, domain-specific gaps
- Edge cases not considered?

Missing Evaluation:

- Important metrics not reported?
- No ablation study?
- No real-world testing?

Poor or Missing Error Analysis:

- Do they explain *why* the model fails?
 - Any qualitative analysis missing?
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10. Keywords & Concepts

4 Core Keywords (essence of the paper):

- 1.
- 2.
- 3.
- 4.

Additional Keywords (minimum 10):

- 1.
 - 2.
 - 3.
 - 4.
 - 5.
 - 6.
 - 7.
 - 8.
 - 9.
 - 10.
-

11. Research Gap Identification

What is NOT solved properly in this paper?

- Limitations you can improve
 - Assumptions you can relax
 - Scenarios not tested
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12. Thesis / Project Idea (Derived)

Possible Research Gap → Thesis Direction:

- How can *you* improve or extend this work?
- New dataset / new model / new evaluation / new application

(Example: "This paper uses CNNs on limited datasets. A possible thesis is applying Transformer-based models on larger, real-world data with better error analysis.")

13. Personal Notes

Your Thoughts: